EMERIO
ENGINEERING ▪ SURVEYING ▪ DESIGN

REVIEWED FOR CODE COMPLIANCE
VALIDITY OF PERMIT
The issuance or granting of a permit or approval of plans, specifications and computations shall not be construed to be a permit for, or an approval of, any violation of the provisions of this code or of any other ordinance of the jurisdiction
DATE 1/7/2026
BLDG-2514996

REGISTERED PROFESSIONAL ENGINEER
105687PE
Joshua D Meyer
OREGON
SEPTEMBER 10, 2024
JOSHUA D. MEYER
EXPIRES: 12/31/2026

TECHNICAL MEMORANDUM

Date: December 20, 2024
To: Washington County Planning Department
From: Josh Meyer, PE – Emerio Design
Subject: Preliminary Drainage at Tax Map 2N4W 32S, Tax Lot 0400

This letter is to serve as a drainage narrative to address runoff from the proposed gravel driveway and building at the above listed tax lot, and to analyze the impact of new onsite stormwater infrastructure.

Site Description

The proposed project site is located on the east side of NW Timmerman Road at the intersection with NW Wilson School Road. The property is zoned EFU. The proposed onsite development will consist of a new building and associated gravel driveway. The proposed gravel driveway is 12 feet wide and 729 feet long and placed to follow existing grade ending at the location of the proposed building.

The current site is primarily vegetated with trees. A portion of the gravel driveway and a culvert were recently added to the site. The 6’ culvert conveys runoff from an unnamed tributary stream to Little Beaver Creek that passes through the site. The new gravel driveway will have portions removed and other portions widened to match the final proposed driveway section. The culvert will be retained under the new gravel driveway.

Stormwater Analysis

Runoff from the proposed building and driveway will route to the onsite tributary stream via v-ditch sections that run parallel to the gravel driveway. Roof runoff routes to the adjacent ditch via storm laterals. Both ditch sections follow the gravel driveway profile, and outfall into the tributary stream directly upstream of the new culvert. Ditch side slopes are proposed at a 4:1 grade.

The new onsite culvert is 6’ diameter corrugated metal. The existing upstream basin currently draining to the culvert was determined based on GIS contours which is currently 167 acres of largely rural/farmland. This area was conservatively assumed to be 10% impervious cover while the curve number for the pervious area determined by NRCS soil survey is 74 for class C soil. This was modeled in HydroCAD V.10 using SBUH modeling which yielded a peak flow rate of 52 CFS during the 25-year 24-hour storm event. Using the flow rate and culvert diameter with a culvert nomograph resulted in a HW/D less than 0.5. The additional 0.36 cfs of runoff from new onsite impervious areas has a negligible effect on the utilized culvert capacity.

A section of the unnamed stream upstream of the new culvert was analyzed for capacity to determine the onsite drainage hazard area (DHA). The DHA is equivalent to the 25-year flow depth through the stream. The stream cross section and approximate slope were input into HydroCAD based on site survey and GIS data. It was determined that the DHA runs at a flow depth of 2.30’. The culvert was also modeled to determine its effects on the 25-year water surface elevation. It was determined that flow depths decrease by approximately 0.55’ when passing through the culvert, leaving no negative impact to the stream or upstream properties. See attachments D and E for plots of the HydroCAD modeling and culvert nomograph. The DHA was mapped on the site exhibits in Attachment A. Note that the DHA in areas outside of

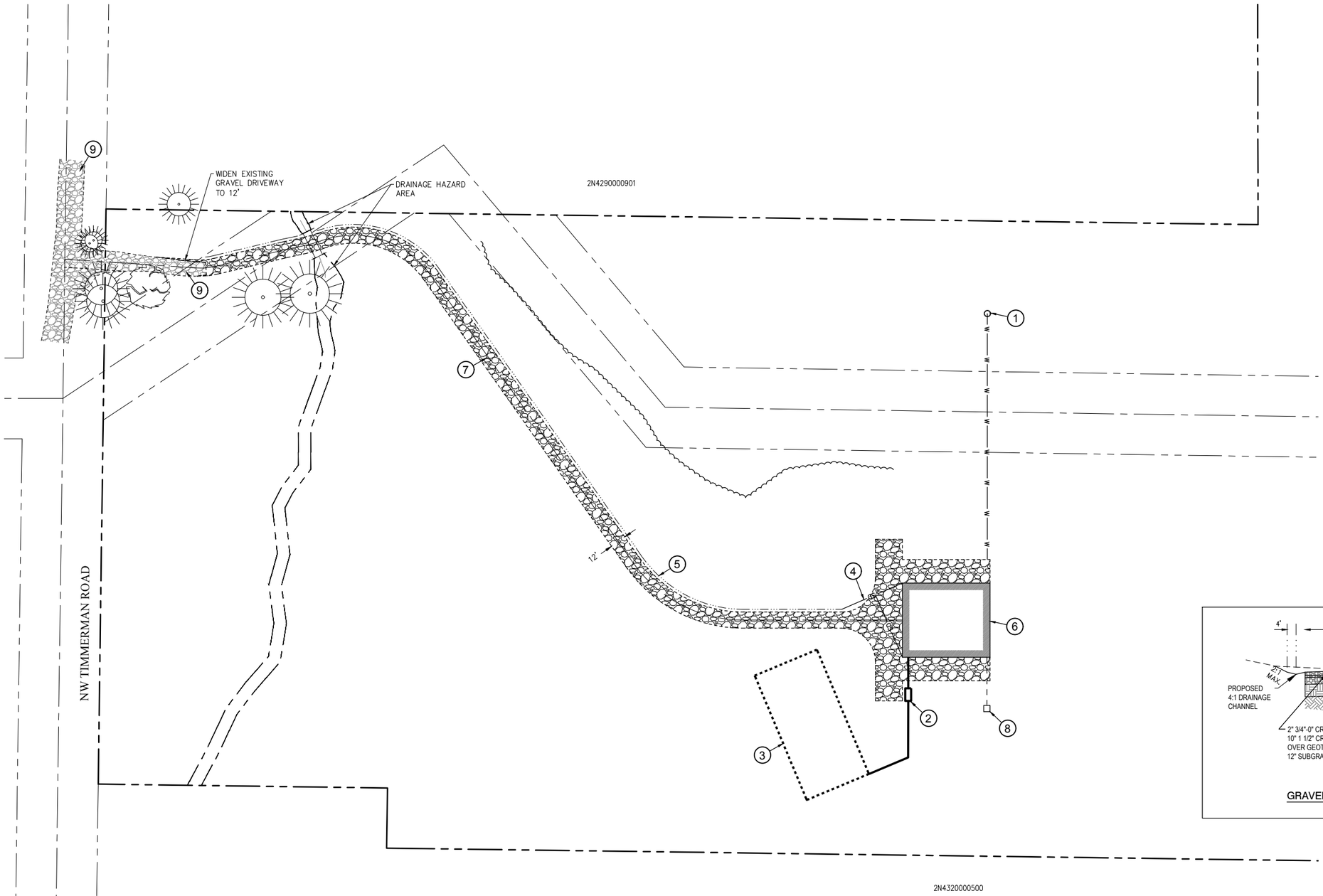
the site survey are approximations based on nearby survey contours and the stream path as determined by GIS data.

Engineering Conclusions

Based on my observations there will be no negative effects to the buildings or other infrastructure onsite or downstream of the site.

List of Attachments:

A. Site Exhibits
B. NRCS Soil Survey – Washington County
C. Upstream Basin Area
D. HydroCAD Output
E. Culvert Nomograph



NW TIMMERMAN ROAD

TAX MAP T2N R4W 32
TAX LOT 400
FOREST GROVE, OREGON

SITE EXHIBIT 1

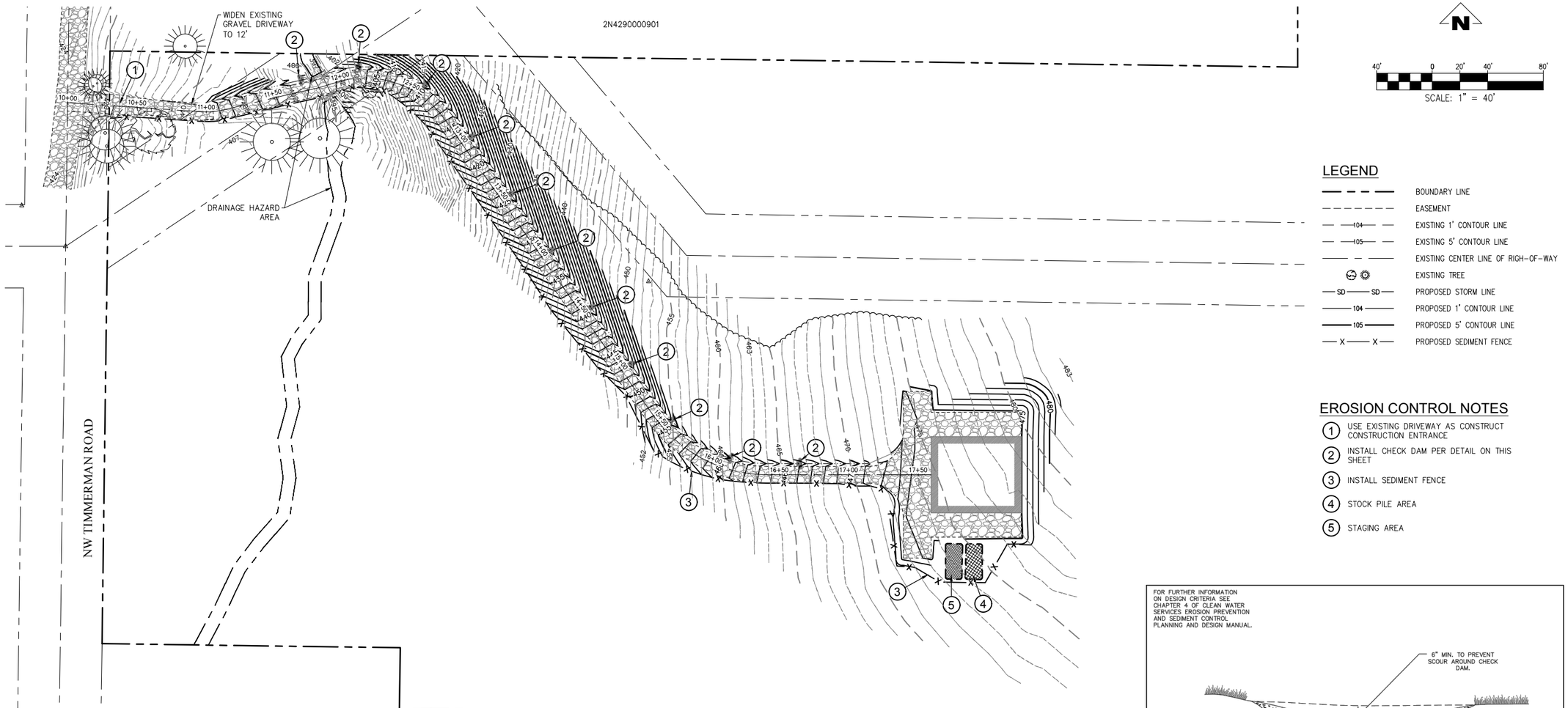
REVISIONS

NO.	DATE	DESCRIPTION

EMERIO
ENGINEERING • SURVEYING • DESIGN
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BEAVERTON, OREGON 97008
TEL: (503) 746-8812
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SHEET 1 OF 2

FILE P: \1916-001 Timmerman Road\docs\civil\storm\1916-001 SITE, Layout: 1 PRELIMINARY SITE & UTILITY PLANS, Plot Date: 12/19/2024 4:13 PM, by: Josh Meyer

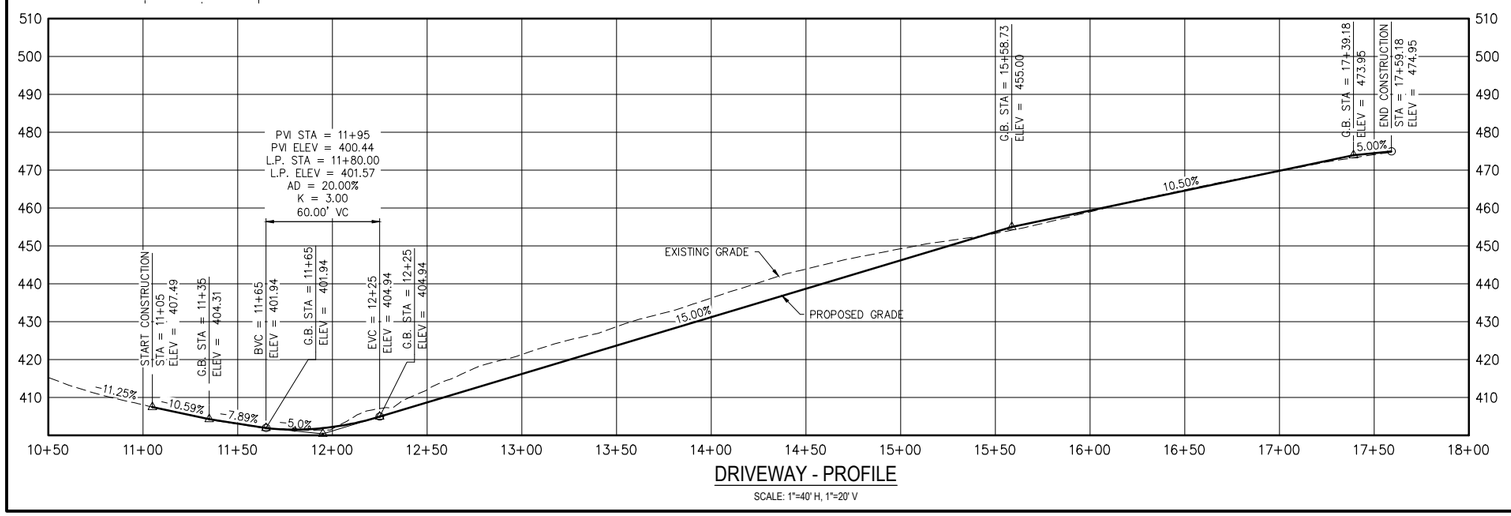


GRADING & EROSION CONTROL PLAN

NW WILSON RIVER HIGHWAY

SCALE: 1"=40'

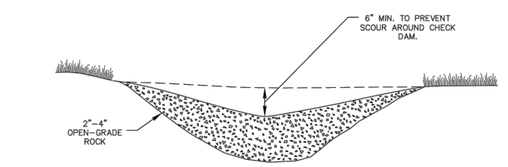
2N4320000500



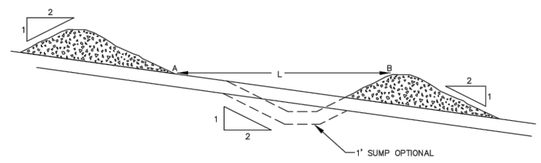
DRIVEWAY - PROFILE

SCALE: 1"=40' H, 1"=20' V

FOR FURTHER INFORMATION ON DESIGN CRITERIA SEE CHAPTER 4 OF CLEAN WATER SERVICES EROSION PREVENTION AND SEDIMENT CONTROL PLANNING AND DESIGN MANUAL.



ROCK CHECK DAM



SPACING BETWEEN CHECK DAMS

NOTES:

1. L = THE DISTANCE SUCH THAT POINTS A AND B ARE OF EQUAL ELEVATION.
2. SEE DRAWING #940 FOR HEIGHT AND SPACING OF CHECK DAMS.

CHECK DAM
ROCK

DRAWING NO. 840

REVISED 10-31-19

Clean Water Services

NW TIMMERMAN ROAD

TAX MAP T2N R4W 32
TAX LOT 400
FOREST GROVE, OREGON

SITE EXHIBIT 2

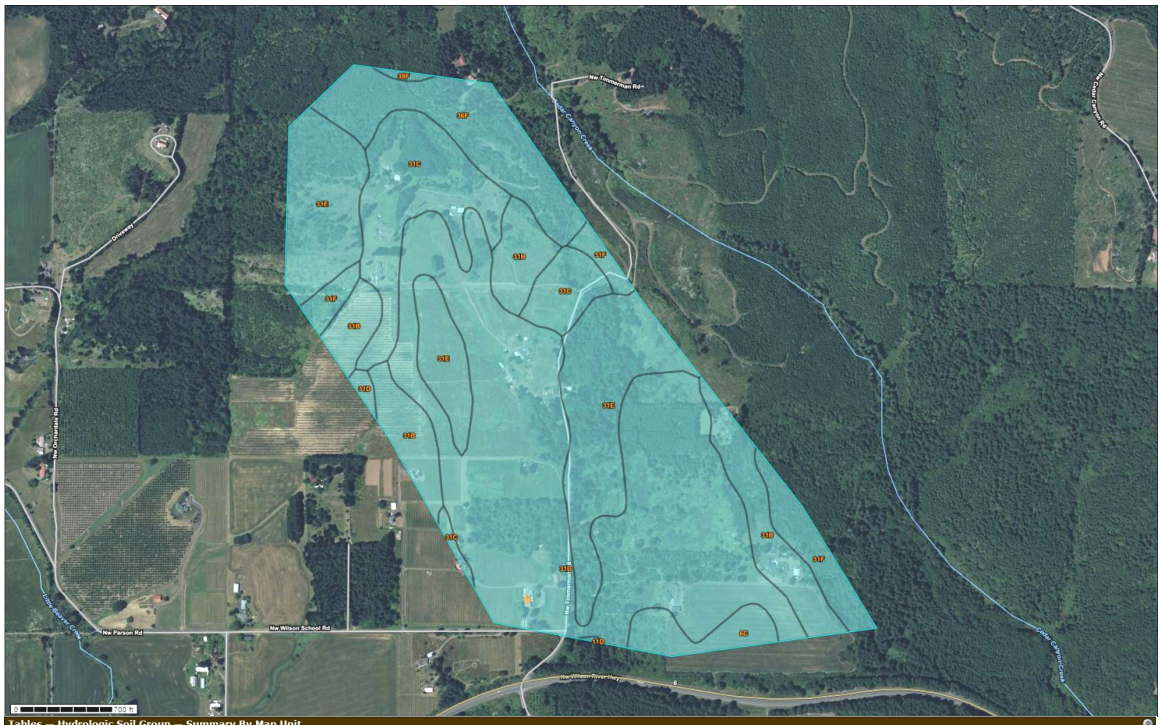
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SHEET 2 OF 2

FILE: P:\1916-001 Timmerman Road\docs\civil\storm\1916-001 GRAD, Layout: 4 PRELIMINARY GRADING & EROSION CONTROL PLAN & DRIVEWAY PROFILE, Plot Date: 12/19/2024 4:14 PM, by: Josh Meyer



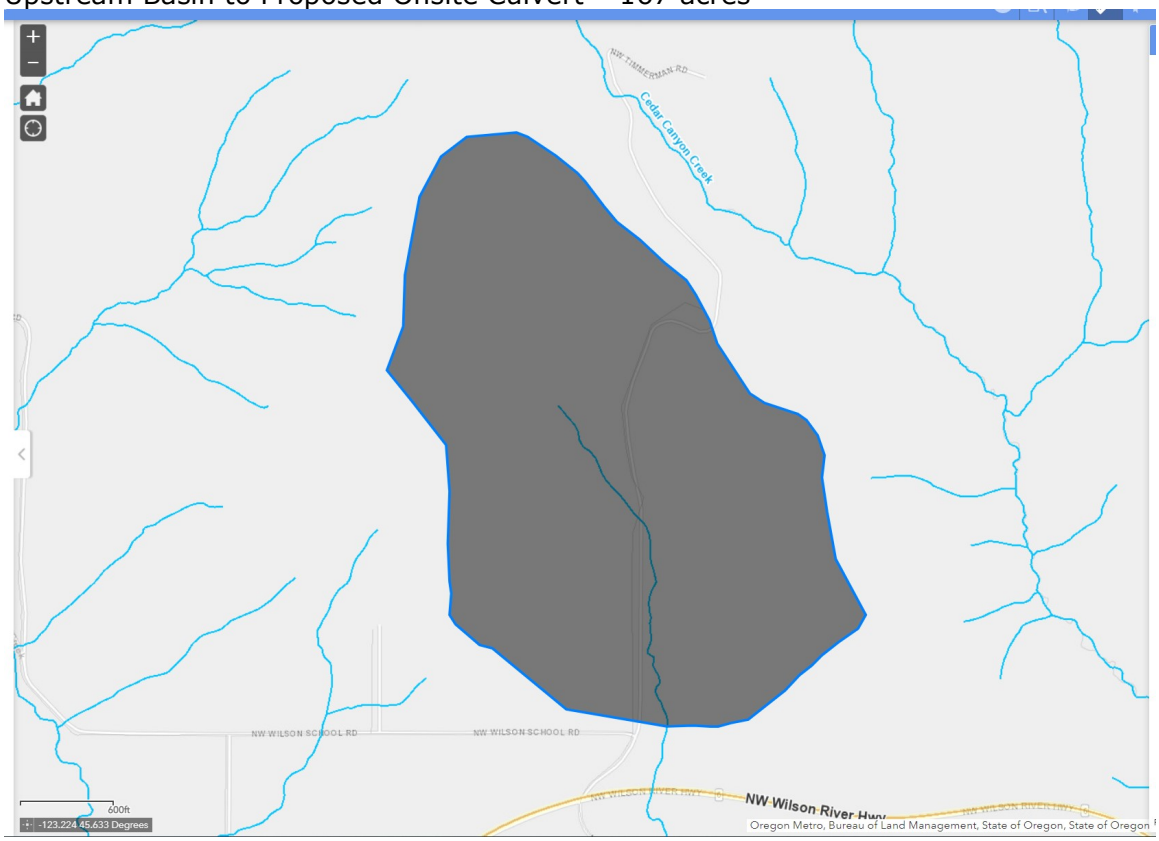
Tables — Hydrologic Soil Group — Summary By Map Unit

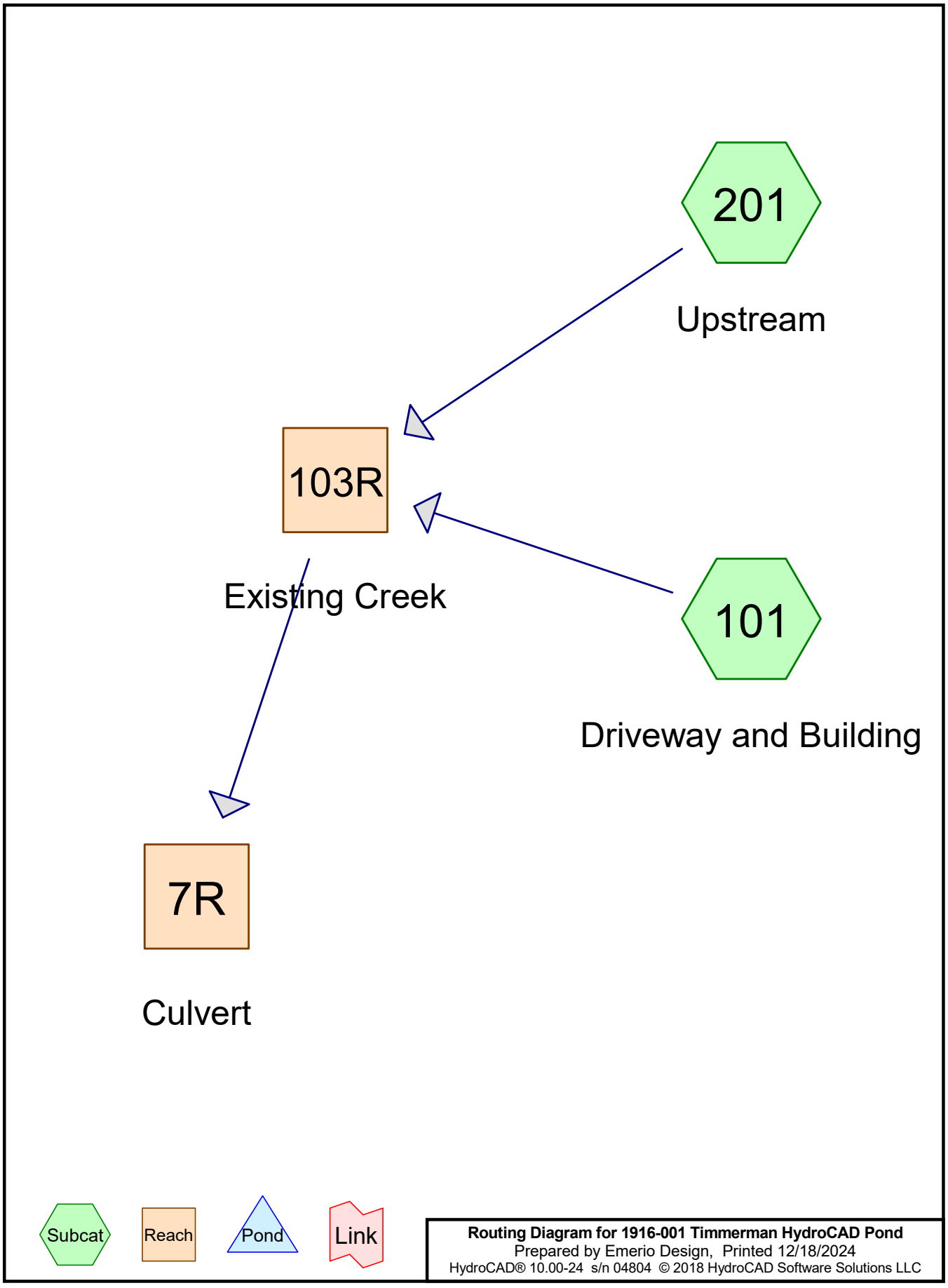
Summary by Map Unit — Washington County, Oregon (OR067)

Summary by Map Unit — Washington County, Oregon (OR067)

Map unit symbol	Map unit name	Rating	Acres in AOI	Percent of AOI
6C	Carlton silt loam, 7 to 12 percent slopes	C	6.3	2.5%
11D	Cornelius and Kinton silt loams, 12 to 20 percent slopes	C	0.1	0.0%
31B	Melbourne silty clay loam, 2 to 7 percent slopes	C	29.1	11.6%
31C	Melbourne silty clay loam, 7 to 12 percent slopes	C	29.8	11.9%
31D	Melbourne silty clay loam, 12 to 20 percent slopes	C	102.5	40.9%
31E	Melbourne silty clay loam, 20 to 30 percent slopes	C	49.3	19.7%
31F	Melbourne silty clay loam, 30 to 60 percent slopes	C	11.7	4.7%
36F	Pervina silty clay loam, 30 to 60 percent slopes	C	21.1	8.4%
38F	Saum silt loam, 30 to 60 percent slopes	C	0.5	0.2%
Totals for Area of Interest			**250.4**	**100.0%**

Upstream Basin to Proposed Onsite Culvert – 167 acres





Routing Diagram for 1916-001 Timmerman HydroCAD Pond
Prepared by Emerio Design, Printed 12/18/2024
HydroCAD® 10.00-24 s/n 04804 © 2018 HydroCAD Software Solutions LLC

Summary for Subcatchment 101: Driveway and Building

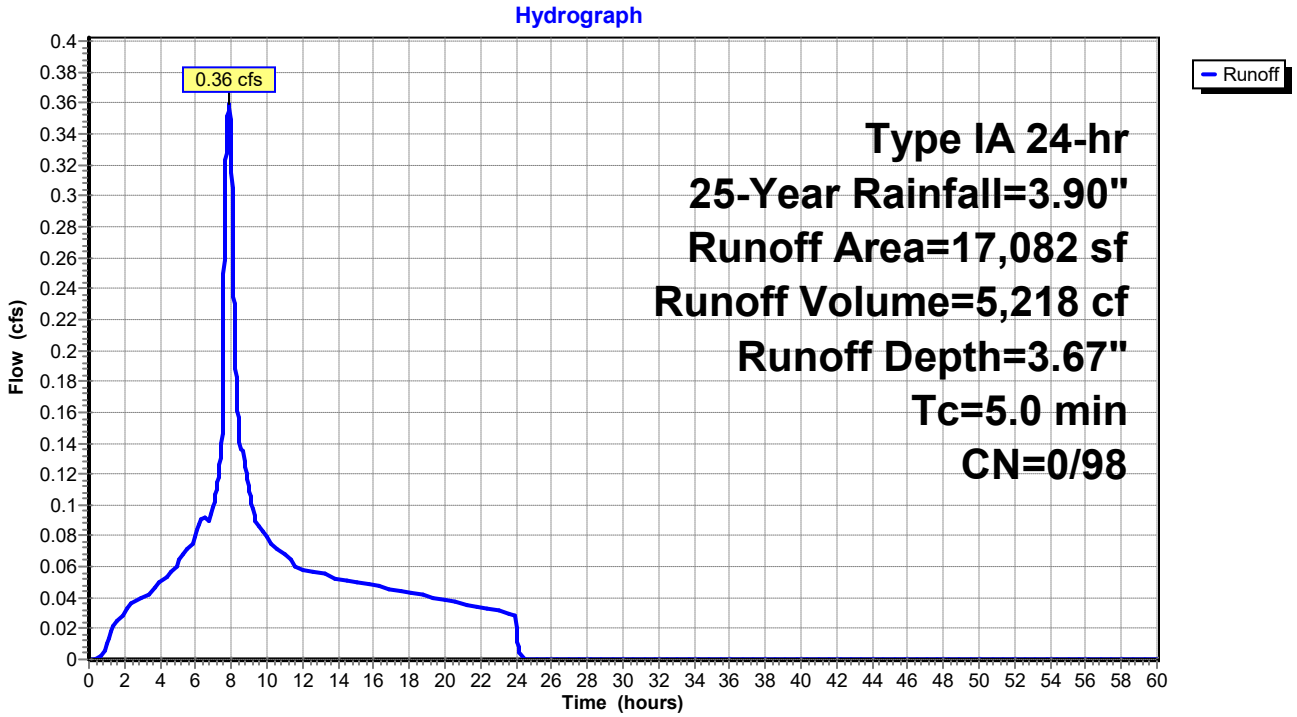
Runoff = 0.36 cfs @ 7.88 hrs, Volume= 5,218 cf, Depth= 3.67"

Runoff by SBUH method, Split Pervious/Imperv., Time Span= 0.00-60.00 hrs, dt= 0.01 hrs
Type IA 24-hr 25-Year Rainfall=3.90"

	Area (sf)	CN	Description
*	17,082	98	Impervious
	17,082	98	100.00% Impervious Area

Tc (min)	Length (feet)	Slope (ft/ft)	Velocity (ft/sec)	Capacity (cfs)	Description
5.0					**Direct Entry,**

Subcatchment 101: Driveway and Building



Summary for Subcatchment 201: Upstream

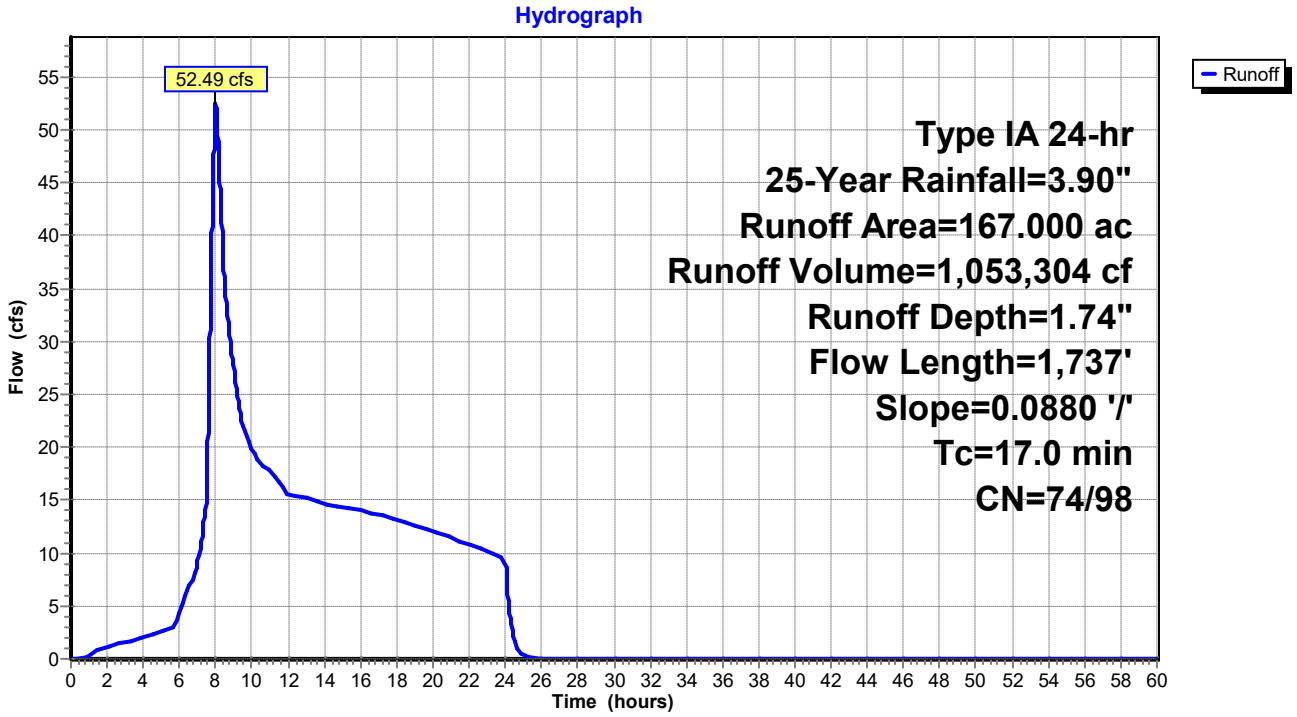
Runoff = 52.49 cfs @ 8.01 hrs, Volume= 1,053,304 cf, Depth= 1.74"

Runoff by SBUH method, Split Pervious/Imperv., Time Span= 0.00-60.00 hrs, dt= 0.01 hrs
Type IA 24-hr 25-Year Rainfall=3.90"

	Area (ac)	CN	Description
*	16.700	98	Impervious
	150.300	74	>75% Grass cover, Good, HSG C
	167.000	76	Weighted Average
	150.300	74	90.00% Pervious Area
	16.700	98	10.00% Impervious Area

Tc (min)	Length (feet)	Slope (ft/ft)	Velocity (ft/sec)	Capacity (cfs)	Description
3.5	50	0.0880	0.24		**Sheet Flow,** Grass: Short n= 0.150 P2= 2.50"
13.5	1,687	0.0880	2.08		**Shallow Concentrated Flow,** Short Grass Pasture Kv= 7.0 fps
17.0	1,737	Total			

Subcatchment 201: Upstream



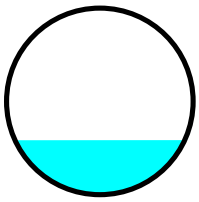
Summary for Reach 7R: Culvert

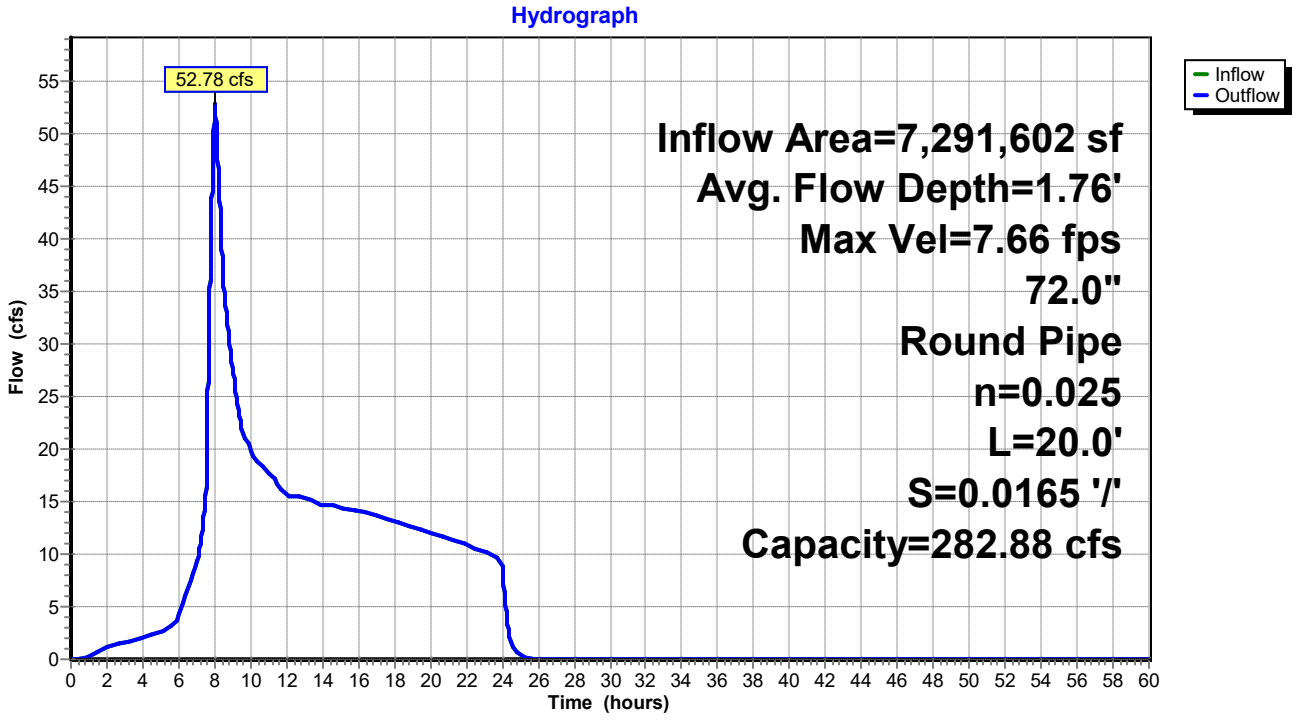
Inflow Area = 7,291,602 sf, 10.21% Impervious, Inflow Depth = 1.74" for 25-Year event
Inflow = 52.78 cfs @ 8.01 hrs, Volume= 1,058,521 cf
Outflow = 52.78 cfs @ 8.01 hrs, Volume= 1,058,521 cf, Atten= 0%, Lag= 0.0 min

Routing by Dyn-Stor-Ind method, Time Span= 0.00-60.00 hrs, dt= 0.01 hrs
Max. Velocity= 7.66 fps, Min. Travel Time= 0.0 min
Avg. Velocity = 4.27 fps, Avg. Travel Time= 0.1 min

Peak Storage= 138 cf @ 8.01 hrs
Average Depth at Peak Storage= 1.76'
Bank-Full Depth= 6.00' Flow Area= 28.3 sf, Capacity= 282.88 cfs

72.0" Round Pipe
n= 0.025 Corrugated metal
Length= 20.0' Slope= 0.0165 '/'
Inlet Invert= 395.64', Outlet Invert= 395.31'





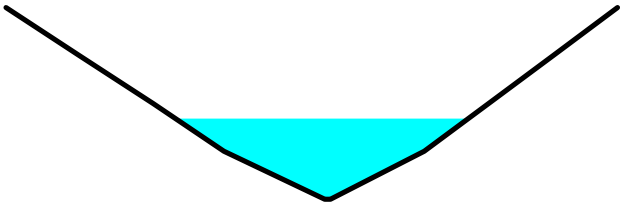
Summary for Reach 103R: Existing Creek

Inflow Area = 7,291,602 sf, 10.21% Impervious, Inflow Depth = 1.74" for 25-Year event
Inflow = 52.83 cfs @ 8.01 hrs, Volume= 1,058,521 cf
Outflow = 52.78 cfs @ 8.01 hrs, Volume= 1,058,521 cf, Atten= 0%, Lag= 0.3 min

Routing by Dyn-Stor-Ind method, Time Span= 0.00-60.00 hrs, dt= 0.01 hrs
Max. Velocity= 9.32 fps, Min. Travel Time= 0.5 min
Avg. Velocity = 5.41 fps, Avg. Travel Time= 0.8 min

Peak Storage= 1,523 cf @ 8.01 hrs
Average Depth at Peak Storage= 1.68'
Bank-Full Depth= 4.00' Flow Area= 27.5 sf, Capacity= 440.42 cfs

Custom cross-section, Length= 269.0' Slope= 0.0640 '/'
Constant n= 0.035 Earth, dense weeds
Inlet Invert= 413.22', Outlet Invert= 396.00'



Offset (feet)	Elevation (feet)	Chan.Depth (feet)
0.00	400.00	0.00
1.53	399.00	1.00
3.07	398.00	2.00
4.55	397.00	3.00
6.65	396.00	4.00
6.77	396.00	4.00
8.73	397.00	3.00
10.07	398.00	2.00
11.42	399.00	1.00
12.76	400.00	0.00

Depth (feet)	End Area (sq-ft)	Perim. (feet)	Storage (cubic-feet)	Discharge (cfs)
0.00	0.0	0.1	0	0.00
1.00	2.2	4.6	578	13.82
2.00	7.7	8.1	2,082	80.63
3.00	16.2	11.6	4,354	216.83
4.00	27.5	15.1	7,400	440.42

Reach 103R: Existing Creek

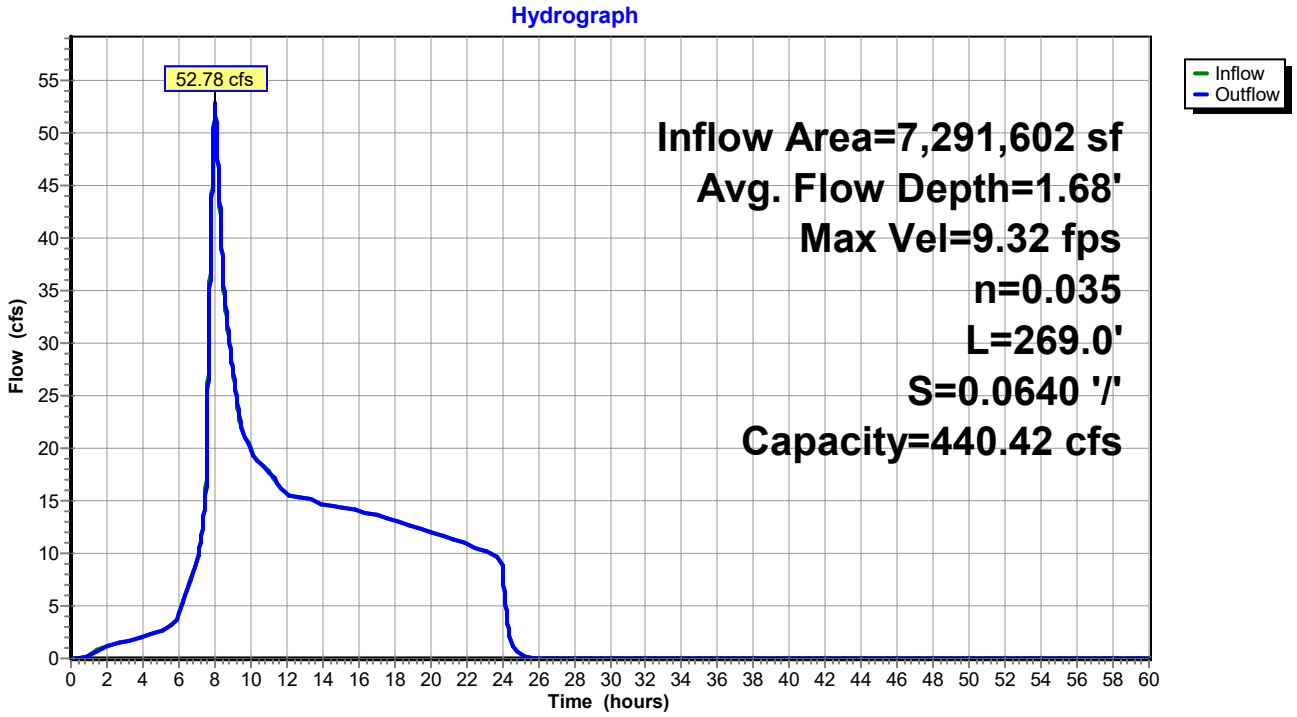
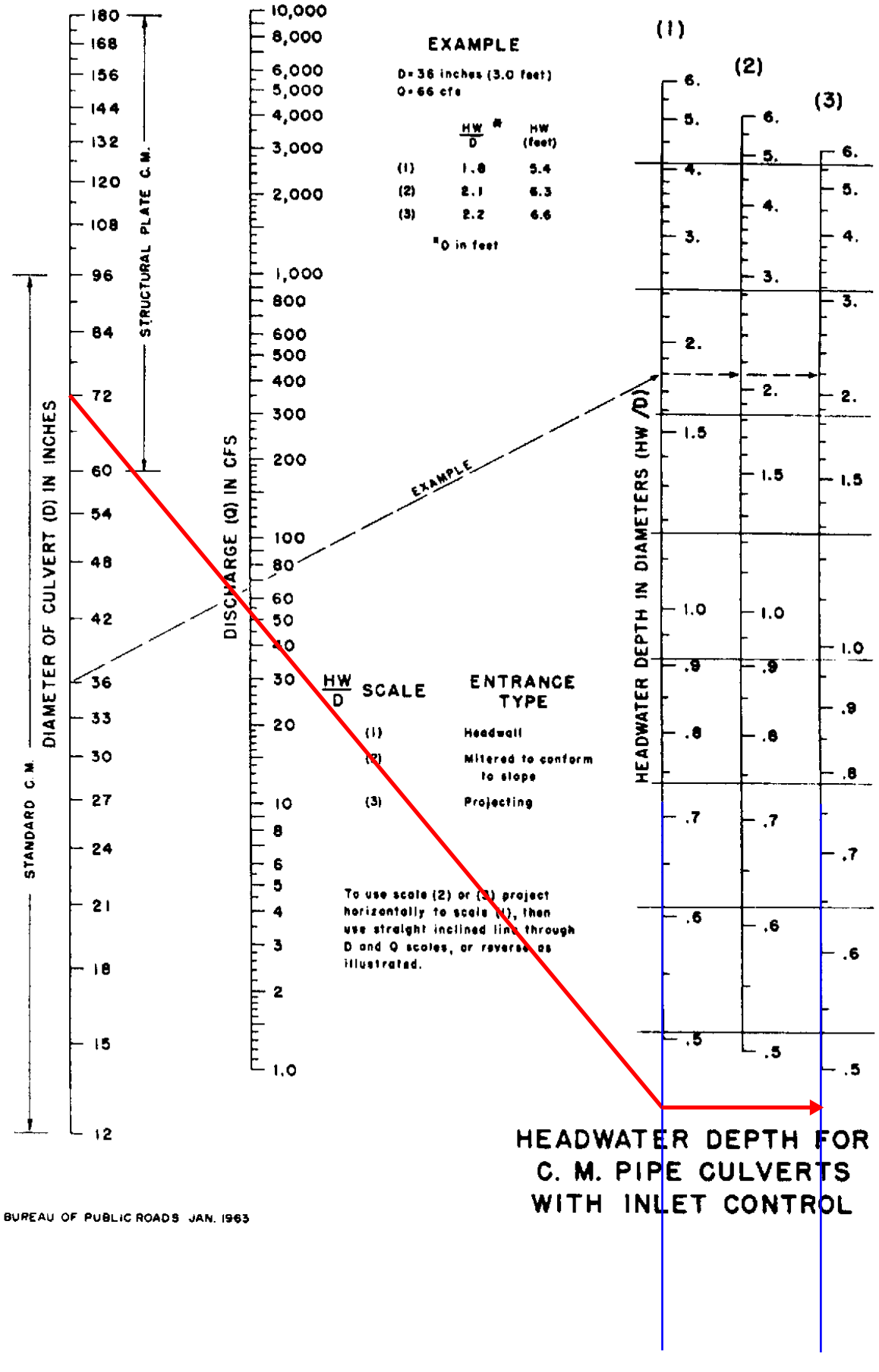


CHART 2B


EXAMPLE
D = 36 inches (3.0 feet)
Q = 66 cfs
HW/D * | HW (feet)
(1) 1.8 | 5.4
(2) 2.1 | 6.3
(3) 2.2 | 6.6
*D in feet
DIAMETER OF CULVERT (D) IN INCHES
STANDARD C.M.
STRUCTURAL PLATE C.M.
DISCHARGE (Q) IN CFS
EXAMPLE
HW/D SCALE
ENTRANCE TYPE
(1) Headwall
(2) Mitered to conform to slope
(3) Projecting
To use scale (2) or (3) project horizontally to scale (1), then use straight inclined line through D and Q scales, or reverse as illustrated.
HEADWATER DEPTH IN DIAMETERS (HW/D)
(1)
(2)
(3)
HEADWATER DEPTH FOR C. M. PIPE CULVERTS WITH INLET CONTROL
BUREAU OF PUBLIC ROADS JAN. 1963